Sanjay Ramkumar

Robotics and Mechatronics | Dynamics Modelling, Control and Optimization



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Education

Present	**Indian Institute of Space Science and Technology** BTech Aerospace Engineering · CGPA - 8.13
Present	**Indian Institute of Technology, Madras** BS Electronic Systems · CGPA - 8.63

Experience

May 2026 – Jul 2026	**U R Rao Satellite Centre (ISRO)** Summer Intern

- Co-designed a two-link lunar monoped (0.2 m jump at 1.625 m/s^2) by nesting CasADi/IPOPT trajectory optimization and a QP task-space controller inside PyGMO differential evolution, with MuJoCo simulations scoring each design on cost of transport.
- Found that the trajectory optimizer beat the QP controller on cost of transport (1.35 vs 1.61) and that each morphology performed best with the controller it was co-designed with.
- Built a 5-DOF extension (base pitch and horizontal translation) to produce stable, self-righting 2D jumps.

Aug 2025 – May 2026	**Small Spacecraft and Payloads Centre** ADCS Team Member

- Designed and simulated attitude control for the InspireSat-3 student CubeSat in a 500 km SSPO, using B-dot detumbling with magnetorquers (settling from a 10°/s tumble in about 58 min about the lower inertia axis).
- Built a 6-DOF Simulink model with PID-controlled reaction wheels for pointing and ground station tracking, and added wheel desaturation logic.
- Redesigned the reference trajectory transitions for first-order smoothness, which reduced torque spikes and improved tracking.

Skills

MuJoCo, MATLAB and Simulink , Python , C++ Programming of Microcontrollers (Arduino, Teensy, ESP) , Robotics , Mechatronics , 2D and 3D design software (Solidworks, AutoCad) , ANSYS

Projects

Jun 2026 – Present	**4 DoF Robotic Arm**

- Designed the arm in SolidWorks across two prototypes, replaced the stock MG995 servo electronics with a custom external PID controller, and built a power distribution board for a 2s2p Li-ion pack.
- Derived closed-form inverse kinematics (validated in Desmos and Simulink) that runs in real time on an Arduino Uno, then moved to MuJoCo with damped least-squares IK.

Apr 2026 – May 2026	**Rotational Imbalance Measurement & Correction System**

- Built a low-cost balancing rig (3D-printed frame, MPU6050 IMU, IR tachometer, Arduino Uno) that finds a rotor's imbalance size and angle from the FFT of its vibration, using baseline and test-mass calibration.
- Located a 1 g imbalance within a few degrees (3.1% average phase error, 12.8% magnitude error) and confirmed the correction mass flattened the FFT peak.

Jun 2024 – Jul 2024	**Two-Wheeled Self Balancing Robot**

- Modeled robot dynamics and designed a cascaded PID controller in Simulink to convert pitch angle into corrective motor torque, validating the control design in simulation before fabrication.
- Built a physical prototype (Teensy, IMU, complementary filter) that recovered from 10° disturbances in under 2 seconds, settling within 10 cm of its original position.
- Engineered a custom wireless PID-tuning tool (NRF24L01, potentiometers) to retune gains live without reflashing firmware.

Competitions

Oct 2025	IN-SPACe CanSat Competition 2025

- Led structural design of a 1 kg, 148 × 242 mm CanSat with a 5-piece ABS isogrid body and spring-hinged deployable rotors, which was designed to handle rocket launch and impact loads, passed a 20 m drop test.
- Built the Simulink model and cascaded PID for the quadcopter secondary descent system and characterised motor thrust on a self-built load cell rig.
- Only team to use a quadcopter for secondary descent.

Jul 2023	RoboCup International 2023 – Bordeaux, France

- Represented India
- Handled electronics, software and control: Teensy 4.0 for low-level motor control, Raspberry Pi 4 running TFLite SSD MobileNet for ball detection (over 85% accuracy), PD speed control per motor and proportional ball tracking.
- Placed **Top 11** and was one of two teams to get every ball into its correct zone.

Nov 2019	RoboCup Asia-Pacific 2019 – Moscow, Russia

- Programmed the line following and ball sorting algorithm for an autonomous rescue robot that handled ramps and obstacles. Won the **Best Technical Interview Award**.

Sep 2019	World Robot Olympiad (WRO) 2019

- Wrote the software for a fully autonomous LEGO robot that follows lines and walls and sorts bulbs by color. Won **First Place at the regional level**.